ADVANCED RESEARCH COMPUTING & DATA AT USC

CENTER FOR ADVANCED RESEARCH COMPUTING (CARC)

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ABOUT CARC

https://www.carc.usc.edu/

The computational expertise in high-performance computing of the USC Center for Advanced Research Computing (CARC) has been a vital resource in USC research community and contributes to improved research productivity and superior outcomes, driving USC's research excellence forward.

CARC aims to be an institutional resource enabling scientific breakthroughs at scale.

Main Service Areas:

1. Advanced Computing
2. Data Solutions
3. Research Support
4. Education

CARC HIGHLIGHTS

The resources, services, and achievements that set us apart.

Advanced Cyberinfrastructure:

- Discovery shared HPC cluster system & Endeavour Condo cluster program
- Artemis: virtual computing platforms and cloud solution (NSF Funded)
- High-performance HPC network upgrade to 200Gbps
- 10+PB data storage capacity

Education & Outreach:

- More than 25 workshop classes and summer bootcamp
- TAC 450: High-Performance Computing for Applied Machine Learning
- NSF CyberTraining project ($300K) – collaboration with AME faculty for development of computational science coursework at Viterbi

Research Collaborations:

- NSF Campus Compute ($400K) - Hybrid Computing System Development
- NSF Regional Network ($1M) - Science DMZ R&E Network Deployment
- NSF Regional Computing ($1M) – Leading Research Computing Alliance in SoCal
- Cryo-EM project for Dornsife & Viterbi –Development of Computational Ecosystem

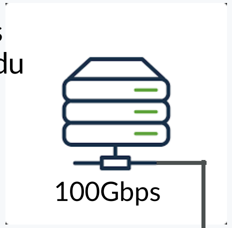
Industry Partnership:

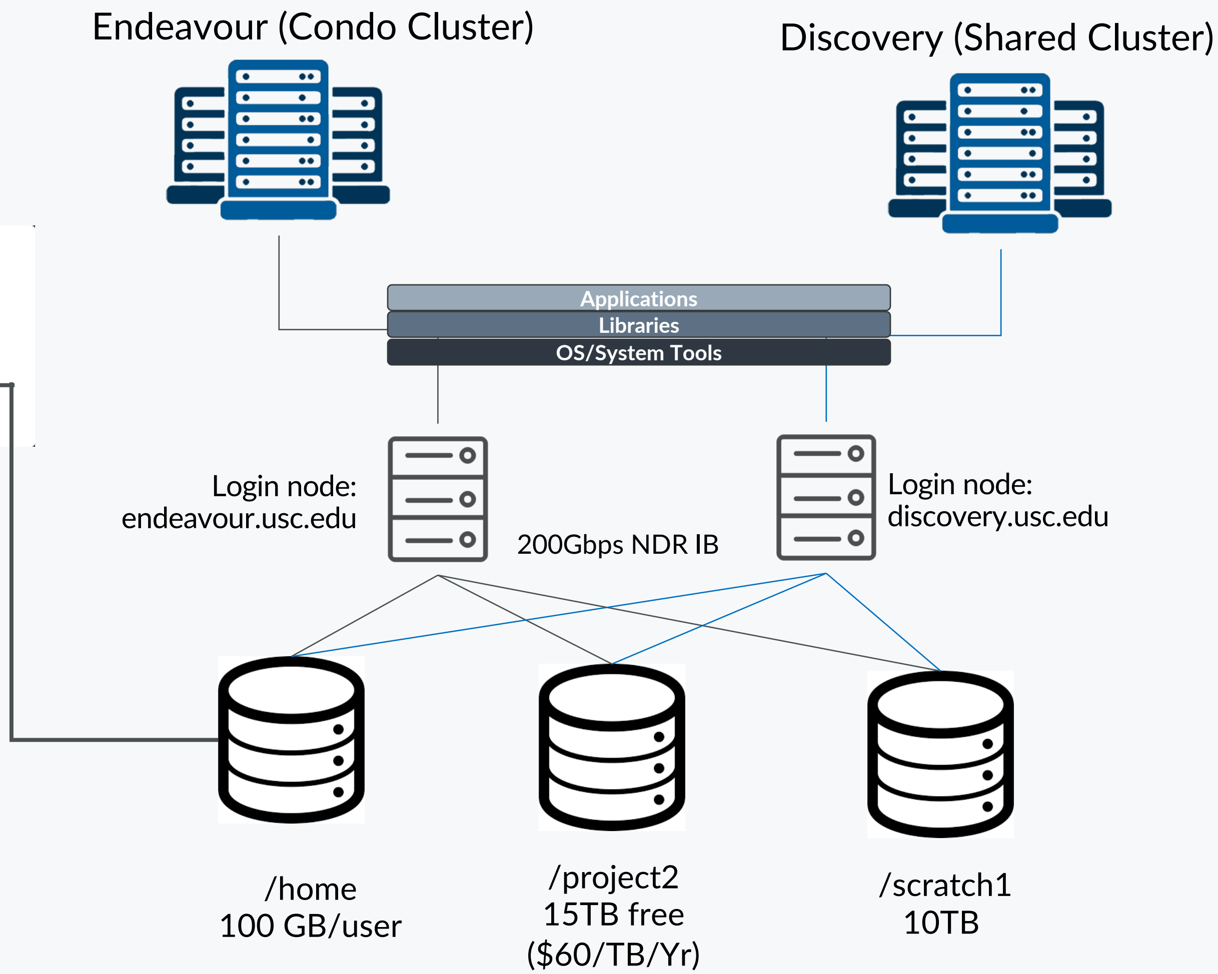
- Samsung Semiconductor, Inc. – Smart Memory, Full NVMe storage solution
- VAST Data – Advanced FS testbed
- Nvidia – Early access to Grace-Hopper next-gen superchip, future collaboration for Frontiers of Computing

CARC SYSTEMS OVERVIEW

CARC systems include the Endeavour condo cluster as well as the Discovery shared cluster

Data Transfer Nodes
hpc-transfer1/2.usc.edu





Home directory (NFS):
Highly Available, not fast
- Installing packages and software
- Storing Keys, Config files, etc.

Scratch directory (PFS):
Very fast, not persistent
- Massive data generated during simulations

Project directory (PFS):
Very fast, persistent
- Long-term storage for research project data

Artemis: Private Cloud
- Virtual computing platform
- DB, Web server, Docker, etc.
- In production since 09/23

Data Archiving Tape System
- 20PB capacity
- Expandable up to 200PB

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HPC **RESOURCES**

The following table summarizes the resources at CARC.

Category	Function	Descriptions
Discovery general-use cluster	Login nodes	2 x 40 Gbps nodes
	Data transfer nodes	2 x 100 Gbps nodes running GlobusConnect
	Compute nodes	~500 nodes, totaling ~18,000 cores
	GPUs	~260 GPUs (A100, L40s, A40, V100, P100)
	Large memory nodes	4 nodes with 1 TB of memory
Endeavour condo cluster	Login nodes	2 x 40 Gbps nodes
	Compute nodes	~900 nodes, totaling ~22,000 cores
	GPUs	~180 GPUs (V100, A40, A100, L40s)
Network	Interconnection	InfiniBand NDR (200 Gpbs)
	Science DMZ	DTN w/ perfSONAR
Storage file systems	/home1	280TB total, 100GB/user ZFS/NFS parallel file system
	/project2	Vast project file system
	/scratch1	Intended for short term compute
Artemis cloud platform	Compute nodes	14 nodes, totaling 896 cores
	GPUs	6 x A40 GPUs

Recognizing the need for a significant upgrade in USC's current HPC infrastructure to meet the demands of future research, CARC is actively engaged in collaborative efforts with multiple departments across the university, focusing on planning and developing next-generation HPC systems that will deliver the necessary computing power to propel USC's research endeavors forward. These concerted efforts aim to provide optimal support for the the Frontiers of Computing initiative, ensuring that USC remains at the forefront of cutting-edge research and innovation.

CCP: CONDO CLUSTER PROGRAM

The Center for Advanced Research Computing (CARC) launched the Condo Cluster Program (CCP) in December 2020 to allow researchers a flexible way to purchase computing resources for their own dedicated use.

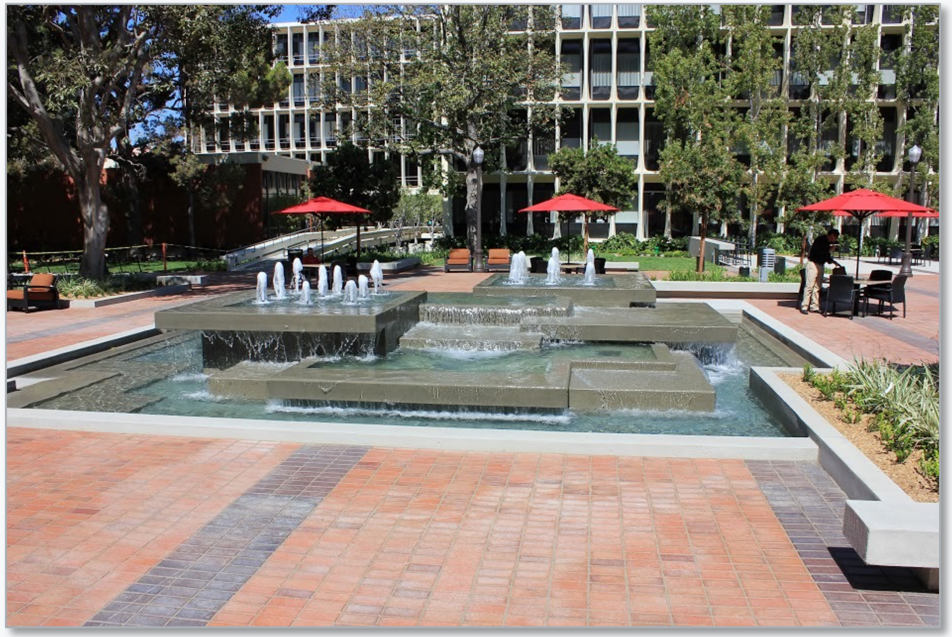
The CCP has two pricing models:

Annual Subscription Model

- Allows research groups to subscribe to their selected number of compute resources on a yearly basis
- Compute nodes can be requested via CARC User Portal
- Allocated nodes get provisioned automatically within a week

Traditional 5-year System Purchase Model

- A useful option when research groups need to make a bulk purchase using a research grant or departmental budget
- Compute/GPU system configurations by CARC
- System purchases can be requested via CARC User Portal



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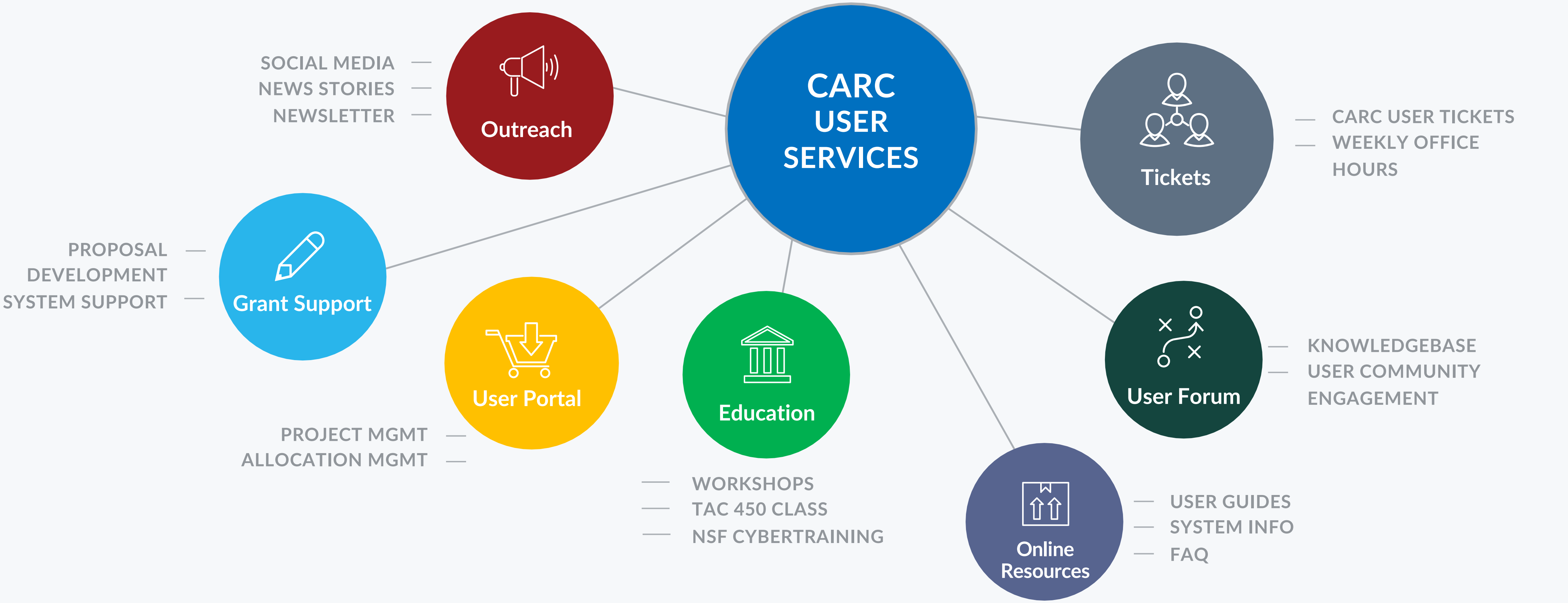
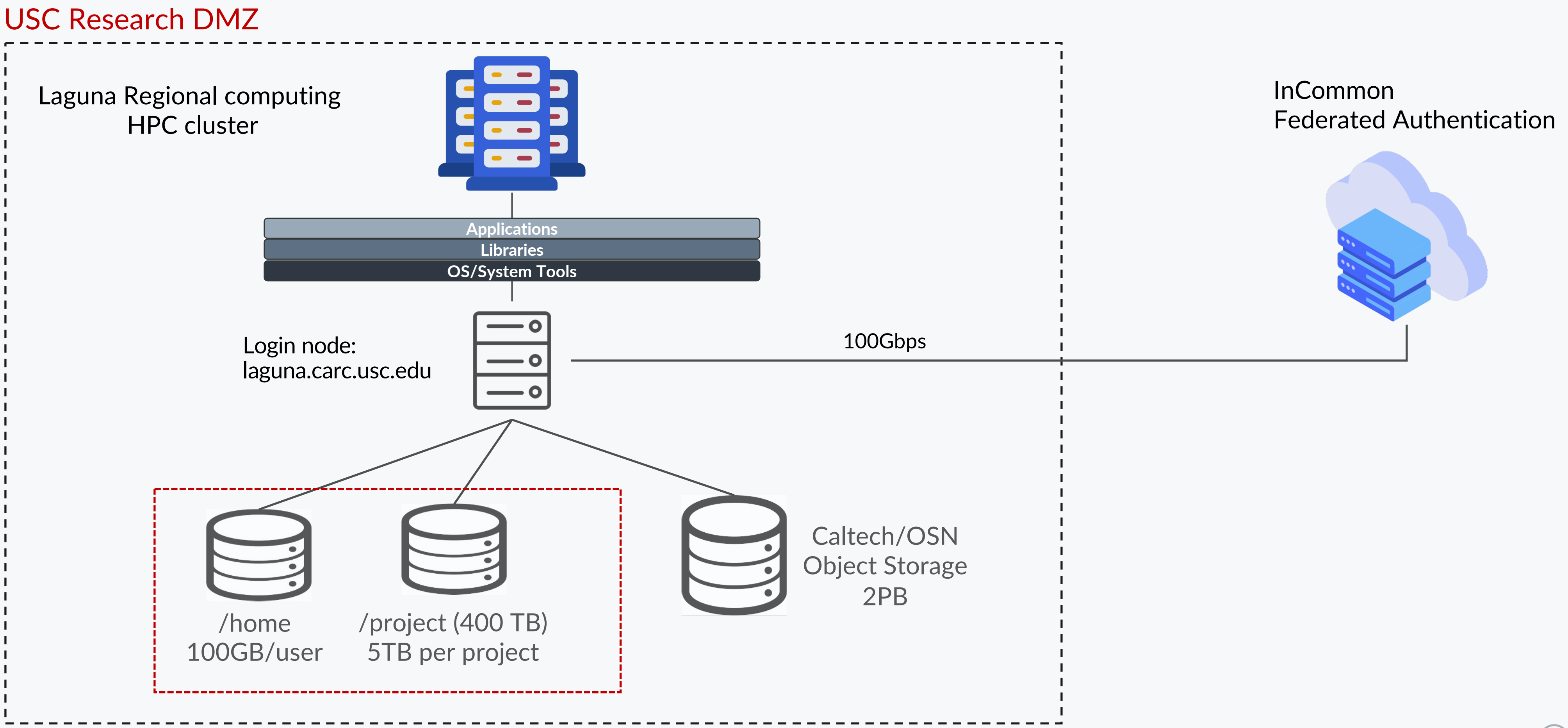
ADVANCED RESEARCH COMPUTING USER SERVICES

The Center for Advanced Research Computing (CARC) offers comprehensive user support services



https://www.carc.usc.edu/ (New CARC Website)

LAGUNA SYSTEM OVERVIEW



LAGUNA: REGIONAL COMPUTING SYSTEM

Dell HPC system configuration:

Category	Function	Qty	Notes
Public facing	Login node	1	Dual AMD EPYC 9354 3.25GHz (32C/64T) w/ 384 GB
Computing	Compute node	16	Dual AMD EPYC 9554 3.1GHz (64C/128T, 256M Cache) CPU 384 GB memory per node
	GPU node	8	Dual EPYC 9354 3.25GHz (32C) + Dual Nvidia L40s GPU (18K CUDA cores, 568 Tensor cores, and 48GB GDDR6 mem) w/ 768GB mem
Network	Interconnection	N/A	Mellanox CX-7 InfiniBand NDR (200 Gbps)
Storage	/home & /project	4	Dual AMD Epyc 9224 2.5GHz (24C/48T) w/192GB memory 12 x 15TB NVMe SSD (4 x 180TB = 720TB)

System installation update:

- System installation completed.
- Home FS is being implemented.
- NVMe storage system (project FS) is installed.
- OSN storage system has been installed, configuration in progress

INCOMMON FEDERATED AUTHENTICATION

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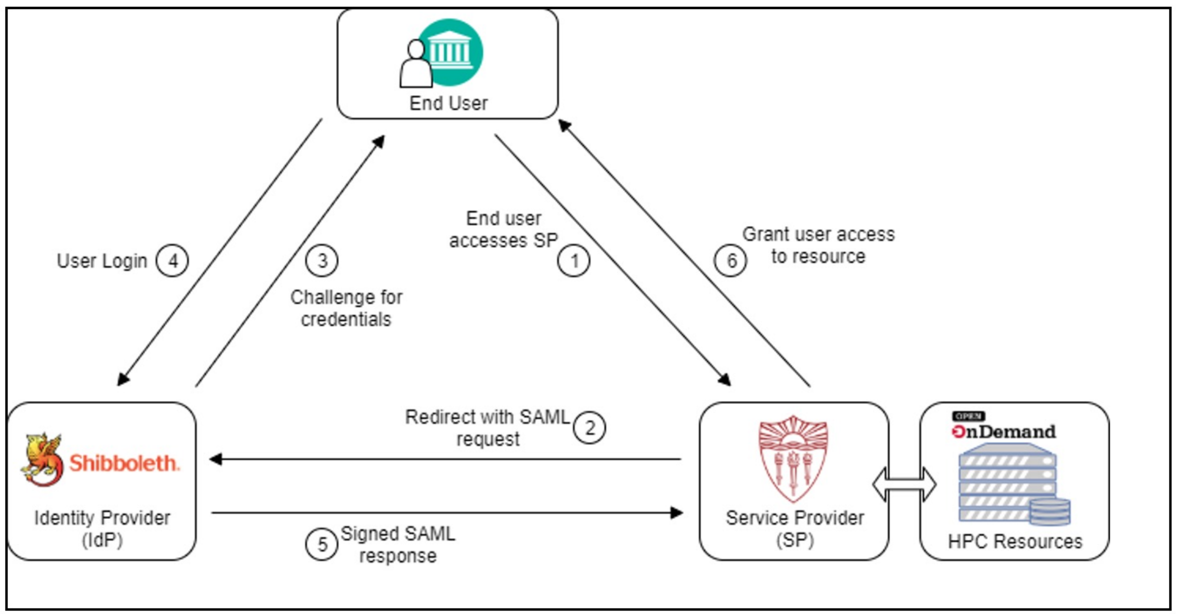

- InCommon Service Provider registration. Registered CARC Shibboleth SP and also for the "Research and Scholarship" category to streamline federation research access.
- USC SSO implementation - Deploy InCommon SSO on CARC endpoints.

THANK YOU & FIGHT ON!

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